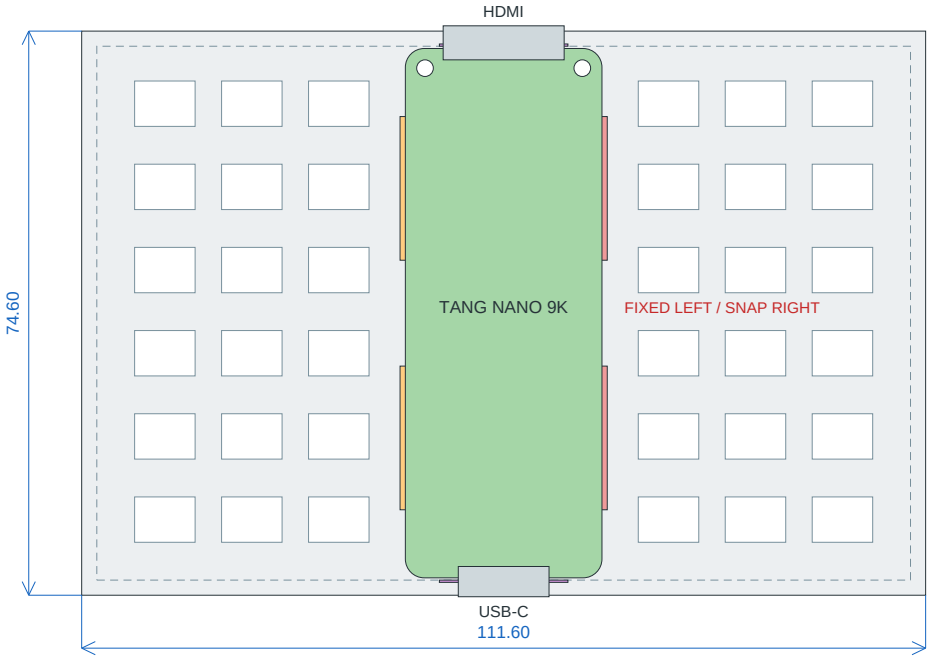
Tang Nano 9K + 4.3-inch LCD panel case

Assembly placement — rear-facing views at true 1:1 scale

SHEET 1/10

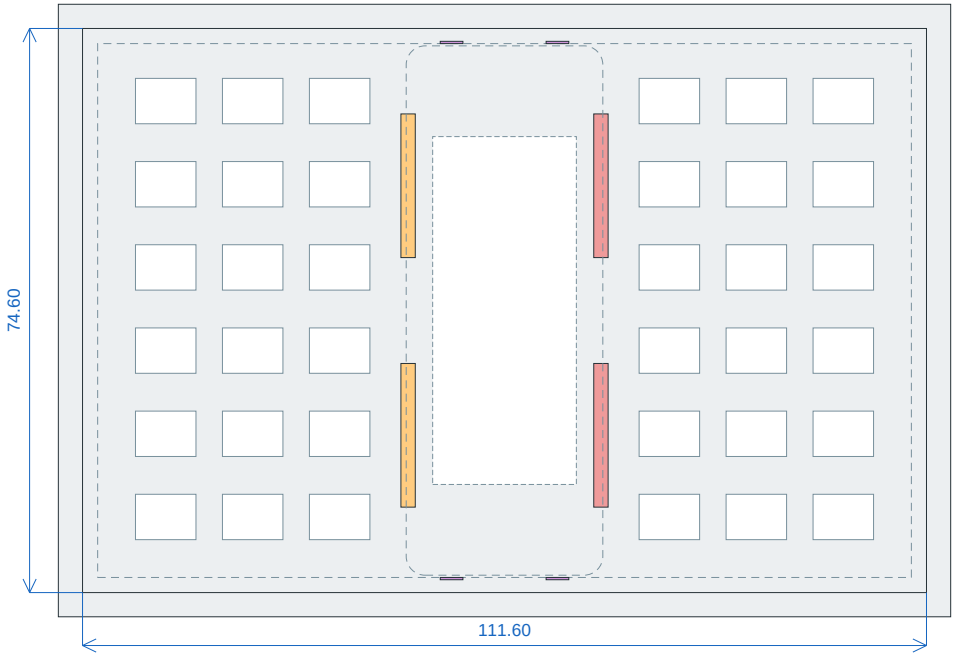
UNITS mm SCALE 1:1

STEP 1 INSTALL PCB IN REAR-COVER RAILS



Component side faces LCD; microSD side faces rear service aperture.

STEP 2 FLIP COVER AND SNAP INTO CHASSIS



Dashed: Tang Nano 9K behind rear cover

HDMI = TOP / USB-C = BOTTOM / microSD = REAR APERTURE

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Printed part A: front chassis

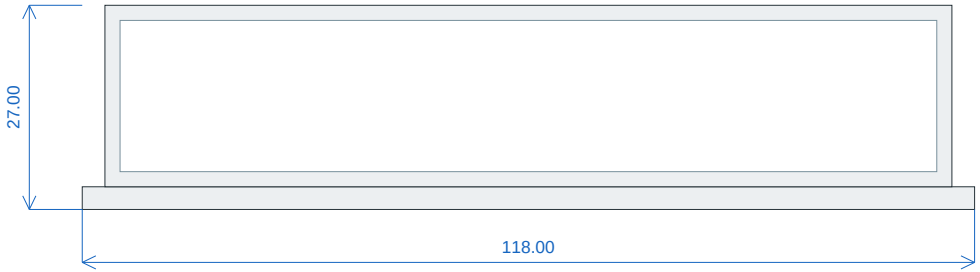
Front outline, centre section, and panel cutout — true 1:1

SHEET 2/10

UNITS mm SCALE 1:1

PANEL CUTOUT 112.60 x 75.60

CENTRE SECTION X-Z / CHASSIS DEPTH 27.00



STL variants: 1.5 / 2.0 / 3.0 mm panels. Outer geometry is common; snap catch depth changes.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

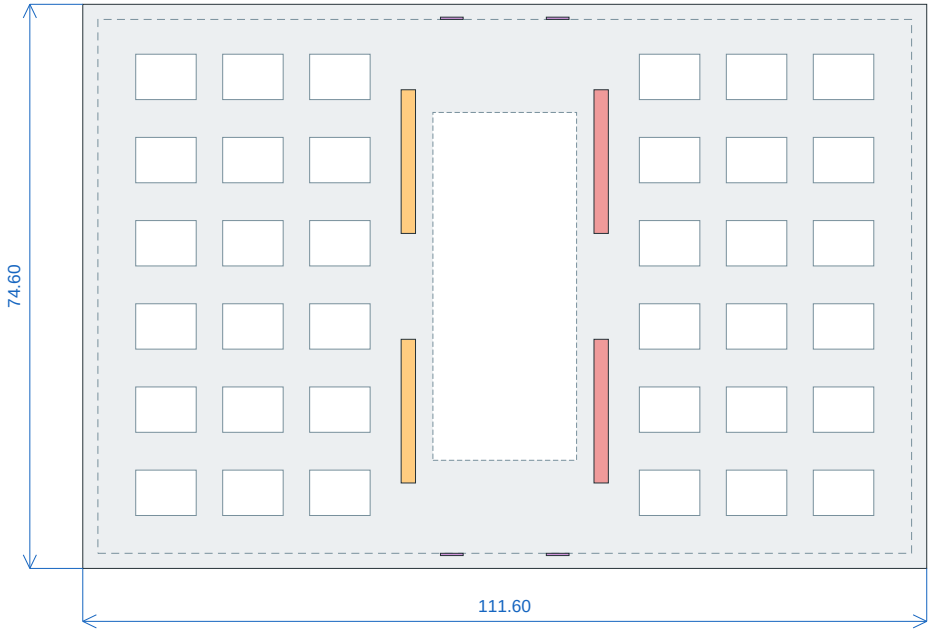
Printed parts B/C

LCD retainer and rear cover with integrated PCB carrier — true 1:1

UNITS mm SCALE 1:1

B LCD RETAINER

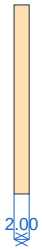
C REAR COVER + PCB CARRIER



Orange: fixed guide. Red: two flexible PCB snap clips.

Centre aperture: microSD access, ventilation, and board removal.

Purple: end stops resist USB-C/HDMI insertion loads.



RETAINER THICKNESS

COVER PLATE THICKNESS

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

CALIBRATION 100.0 mm

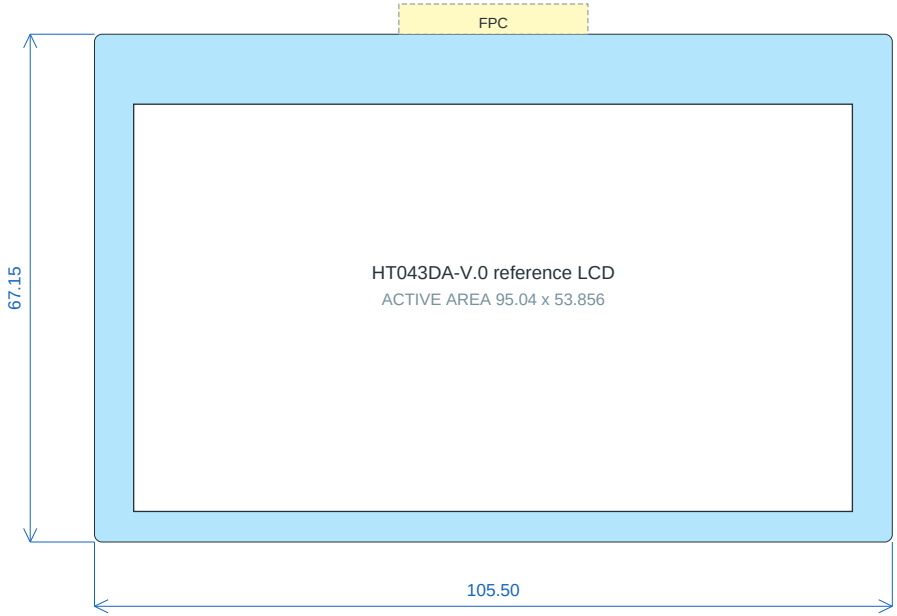
Physical hardware check templates

LCD, Tang Nano 9K, and panel cutout — all true 1:1

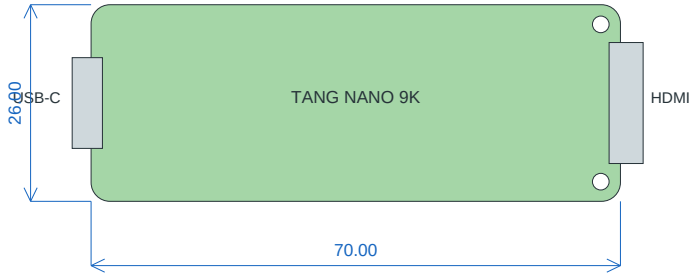
SHEET 4/10

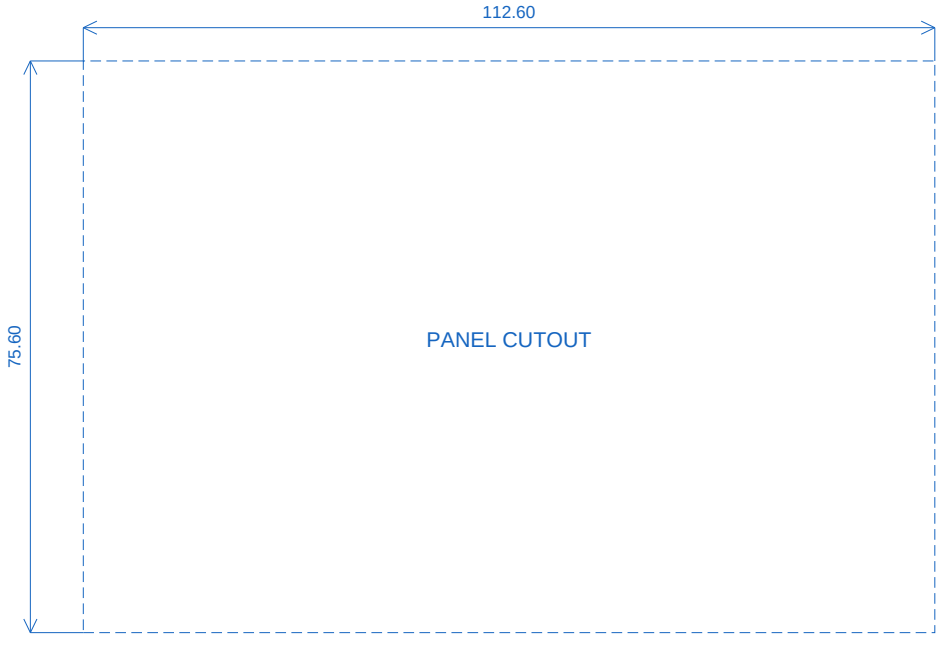
UNITS mm SCALE 1:1

LCD REFERENCE



TANG NANO 9K PCB





CALIBRATION 100.0 mm

DIMENSION STATUS

VERIFIED: PCB outline 70.00 x 26.00 mm

VERIFIED: LCD outline 105.50 x 67.15 x 2.90 mm (HT043DA-V.0)

DESIGN VALUE: panel cutout 112.60 x 75.60 mm

CHECK HARDWARE: connector height/projection and LCD FPC shape

Do not print the STL if your LCD does not match this template.

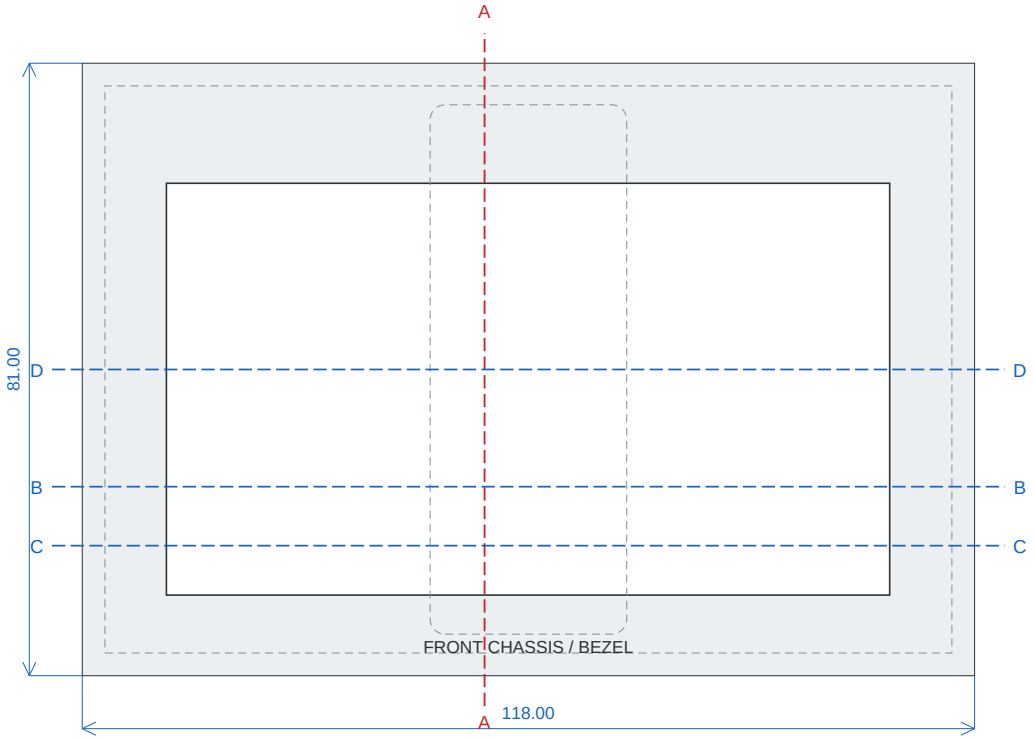
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Assembly section index

Cutting planes are positioned to pass through the actual hooks, clips, connectors and service aperture

UNITS mm SCALE 1:1

REAR PROJECTION / SECTION LOCATIONS / SCALE 1:1



SECTION PURPOSE

A-A LONGITUDINAL Y-Z SECTION

USB-C / HDMI, PCB end stop, LCD, FPC route and rear cover

B-B PCB RETENTION X-Z SECTION

fixed lip, support shelf, PCB and flexible clip

C-C LCD HOOK X-Z SECTION

retainer cantilevers, stepped heads and chassis windows

D-D MICROSD SERVICE X-Z SECTION

PCB underside, service aperture and rear clearance

E-E HDMI-END M2 BOSS X-Z SECTION

two HDMI-end mounting bosses, pilot bores, PCB and deep rear shell

- FRONT CHASSIS
- LCD VERIFIED BODY
- LCD RETAINER
- PCB 70 x 26 x 1.6
- REAR COVER / CARRIER
- REFERENCE ENVELOPE — VERIFY ON HARDWARE

SOLID FILLS: evaluated directly from STL CSG geometry.

DASHED ORANGE: electronics envelope only; measure the physical board.

CALIBRATION 100.0 mm

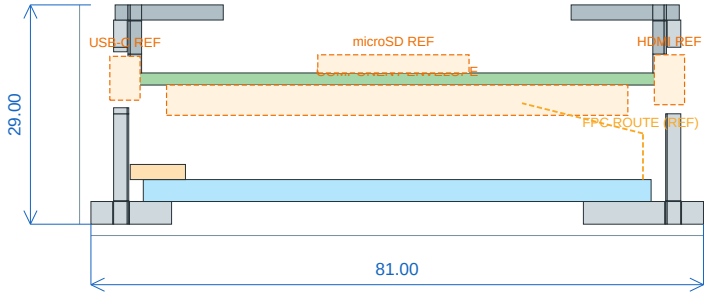
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Exact assembly sections A-A / B-B

Solid geometry is evaluated from the STL source; all mechanical sections are true 1:1

UNITS mm SCALE 1:1

A-A Y-Z @ X=53.20 mm / USB-C → HDMI



WHAT THIS CUT PASSES THROUGH

- verified PCB edge-stop at both connector ends
- USB-C and HDMI wall openings
- LCD body, retainer plate and rear cover
- reference-only connector/component envelopes
- reference FPC routing corridor

FRONT CHASSIS

LCD VERIFIED BODY

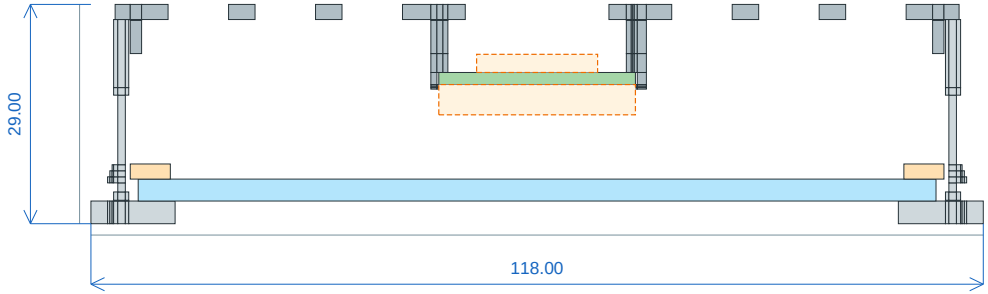
LCD RETAINER

PCB 70 x 26 x 1.6

REAR COVER / CARRIER

REFERENCE ENVELOPE — VERIFY ON HARDWARE

B-B X-Z @ Y=25.00 mm / FIXED LIP ← PCB → FLEX CLIP



The PCB is not floating:

z=18.40 support shelves under both edges

left fixed lip + right 1.20 mm cantilever clip

dashed volumes above/below PCB are unverified components

CALIBRATION 100.0 mm

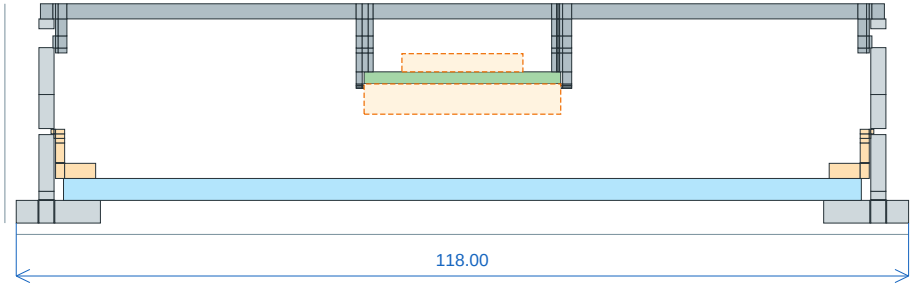
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Exact assembly sections C-C / D-D

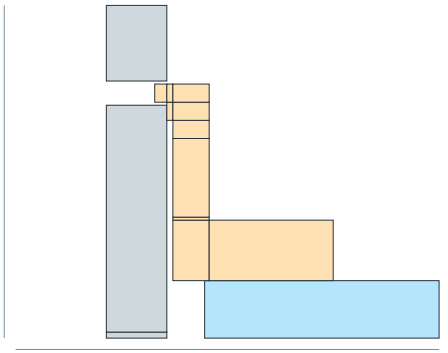
LCD hook engagement and microSD access are shown at their actual cutting planes

UNITS mm SCALE 1:1

C-C X-Z @ Y=17.20 mm / BOTH LCD HOOKS



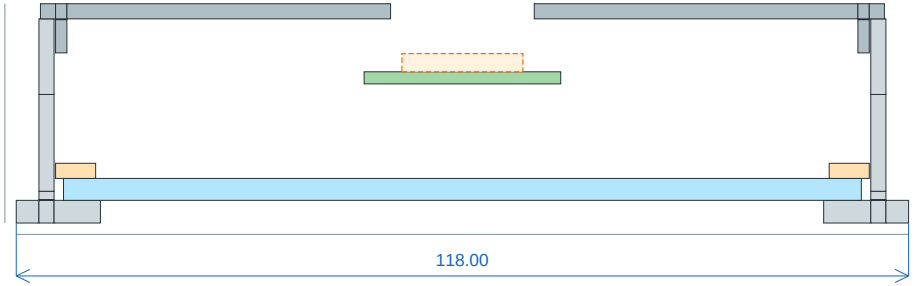
LEFT HOOK DETAIL 4:1



orange: retainer arm/head

grey gap: chassis window z=11.70..12.50

D-D X-Z @ Y=40.50 mm / microSD SERVICE APERTURE



At D-D the rear-cover plate is intentionally absent

under the PCB. This is the service aperture, not a hollow error.

The dashed microSD socket is a reference envelope;

verify its exact location and height on the physical board.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Section interpretation and assembly

The section set distinguishes verified printable geometry from hardware values requiring measurement

UNITS mm SCALE 1:1

ASSEMBLY ORDER

1. Install the selected front chassis in the 112.60 x 75.60 panel cutout.
2. Place the LCD at z=3.00..5.90 with its display toward the bezel.
3. Push the retainer until all four C-C hook heads engage their wall windows.
4. Insert the PCB left edge below the B-B fixed lips and press the right edge into both clips.
5. Confirm the PCB rests on four shelves and between four A-A end stops.
6. Connect the FPC, verify the measured connector/component envelopes, then latch the cover.

SECTION READING RULES

- Filled regions are exact intersections of the generated STL CSG solids.
- Empty regions may be intentional openings: display window, latch window, port opening or service aperture.
- Orange dashed regions are not printable geometry and are not dimensionally certified.
- A-A through E-E coordinates are measured from the bezel left/bottom/front datums.

MEASUREMENT HOLD POINTS

- ☐ USB-C shell: width, height, projection
- ☐ HDMI shell: width, height, projection
- ☐ microSD socket: x/y position and rear height
- ☐ maximum component height on both PCB faces
- ☐ LCD FPC exit point, width and bend radius

Do not treat dashed envelopes as production dimensions.

- FRONT CHASSIS
- LCD VERIFIED BODY
- LCD RETAINER
- PCB 70 x 26 x 1.6
- REAR COVER / CARRIER
- REFERENCE ENVELOPE — VERIFY ON HARDWARE

CALIBRATION 100.0 mm

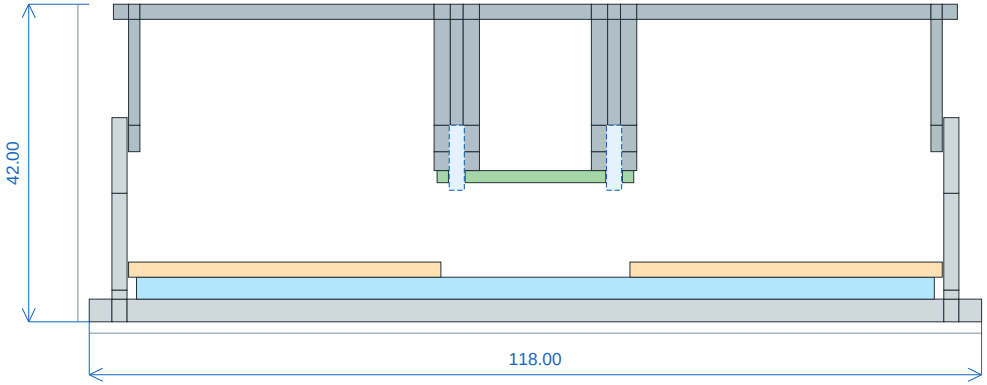
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Expanded rear-cover sections

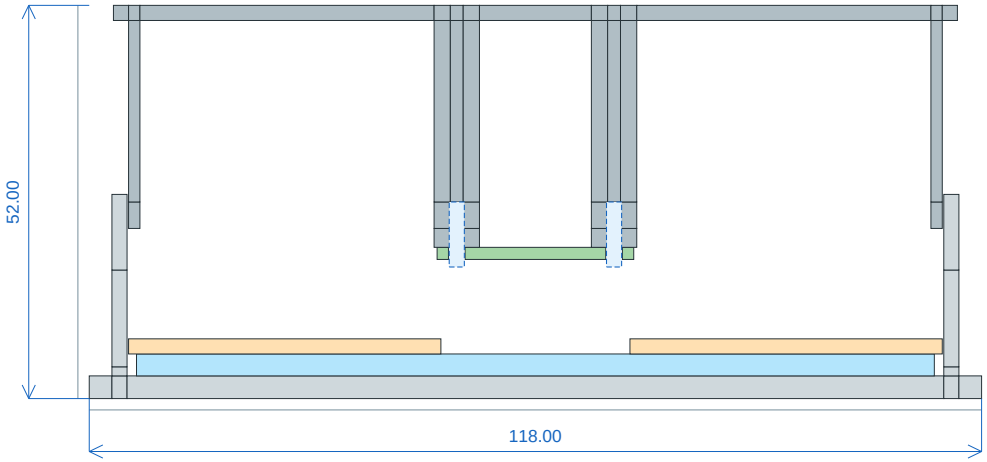
HDMI-end E-E cut through both M2 bosses; PCB and connector planes remain unchanged

UNITS mm SCALE 1:1

20 mm CLEARANCE VARIANT / E-E X-Z / SCALE 1:1



30 mm CLEARANCE VARIANT / E-E X-Z / SCALE 1:1



E-E CUT CONTENT

- two HDMI-end mounting holes only
- two 6.00 mm square printed bosses
- 1.70 x 1.70 mm M2 self-tapping pilot
- 6.00 mm effective thread depth
- optional M2x8 screws from PCB front side
- fixed lip and flex clips remain active

DEPTH DATUM

PCB rear surface: z=20.00 mm in every variant

20 mm cover inner face: z=40.00 mm

30 mm cover inner face: z=50.00 mm

Blue dashed shafts are screw references; grey bosses are STL geometry.

Verify the physical daughterboard outline and cable bend space.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

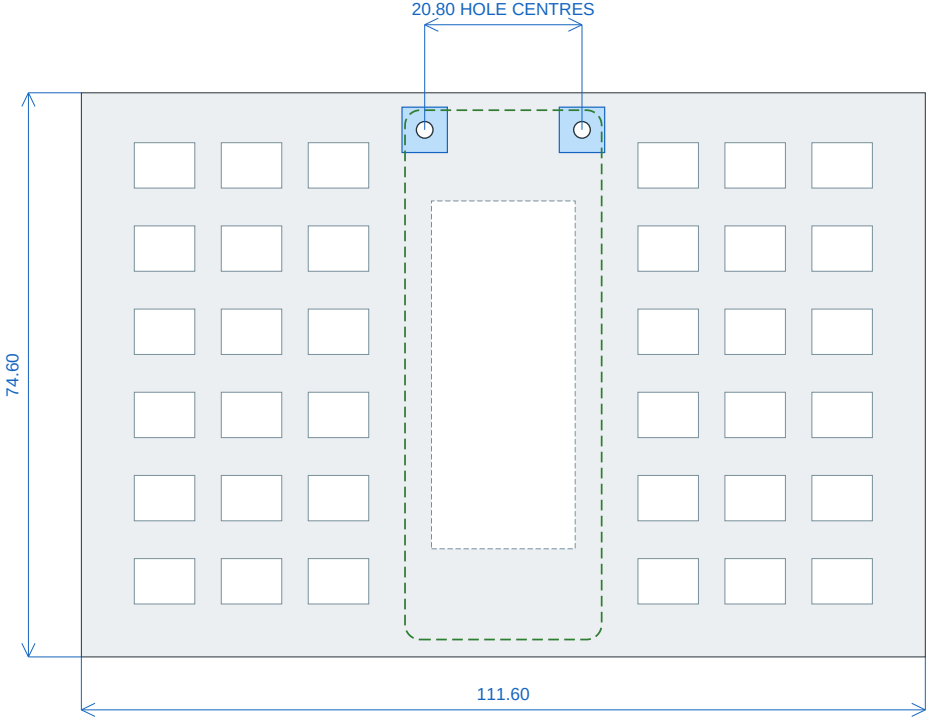
HDMI-end two-hole mounting layout

Rear projection at true 1:1; screw fixation is optional and supplements the snap carrier

SHEET 10/10

UNITS mm SCALE 1:1

REAR-COVER INBOARD VIEW / HDMI END AT TOP



Blue: printed bosses / Green dashed: Tang Nano 9K PCB

MOUNTING SPECIFICATION

PCB holes	2 only, HDMI end
Hole reference	diameter 2.20 mm
Hole-centre spacing	20.80 mm
Printed boss	6.00 x 6.00 mm
Pilot bore	1.70 x 1.70 mm square
Recommended screw	M2x8 self-tapping
Rear clearance variants	20.00 / 30.00 mm
Overall case depth	42.00 / 52.00 mm

ASSEMBLY

1. Insert PCB under the fixed lips and engage both flex clips.

2. Align the two HDMI-end PCB holes with the printed pilots.

3. Install M2x8 screws without compressing or bowing the PCB.

4. Confirm 20/30 mm daughterboard height and cable clearance.

Do not use screws longer than required; the pilot thread depth is 6.00 mm.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.